Aggregated genomic data as cohort-specific allelic frequencies can boost variants and genes prioritization in non-solved cases of inherited retinal dystrophies

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ABSTRACT

We built **an allelic-frequency database** for a heterogeneous cohort of genetic diseases to explore the aggregated genomic information and **boost diagnosis** in inherited retinal dystrophies (IRD).We retrospectively selected 5683 index-cases, including 1766 with IRD. We calculated subcohort's IRD specific allele-frequencies and compare it with suitable pseudocontrols. We **prioritized variants** with a significant increment of frequencies, among them 8 found in IRD non-solved cases that may contribute to explain the phenotype, and **10 out of 11 of uncertain significance that were reclassified as likely-pathogenic** according to ACMG guides. We also developed a method to **highlight genes with more frequent pathogenic variants in non-solved IRD cases** than in pseudocontrols weighted by the increment of benign variants in the same comparison. Thus, we identified 18 genes for further studies that provided **new insights in five cases**. Our resource can also help to calculate the **carrier frequency** of deleterious variants in IRD genes, being the most prevalent ABCA4 (~7%) and USH2A (~3%).

A **cohort-specific genomic database and phenotype-specific allele frequencies** compared to controls can assist with **variants and genes prioritization** and operate as an engine that provides new hypothesis in specific non-solved cases, hence augmenting disease diagnosis rate.

METHODS

Cohort – 5683 cases (2015-2021)

IRD – **N = 1766**
Others – **N = 3917**

Clinical Exome Sequencing

TruSightOne (TSO) – illumina

Clinical Exome Solution (CES) – SOPHiA

Bioinformatics' reanalysis

Pipeline based on GATK 4.1 SNVs caller
https://github.com/TBLabFJD/VariantCallingFJD

Sample duplicates and relatives' detection and removal

plink...

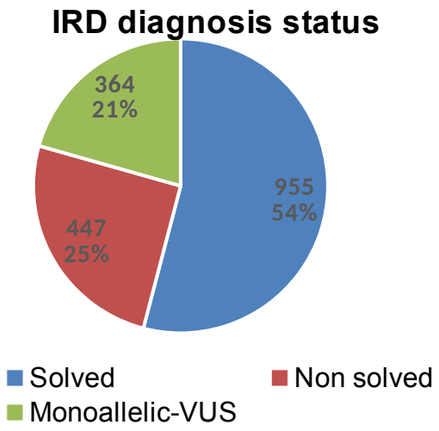
Whole genome association analysis toolset

SUB-COHORTS

IRD N = 1766 — **Disease Specific** frequency

PSEUDOCONTROL N = 3531 — **Pseudocontrol** frequency

IRD diagnosis status



Variant frequency calculation

hail — **1.5 million** unique variants

Gene panels

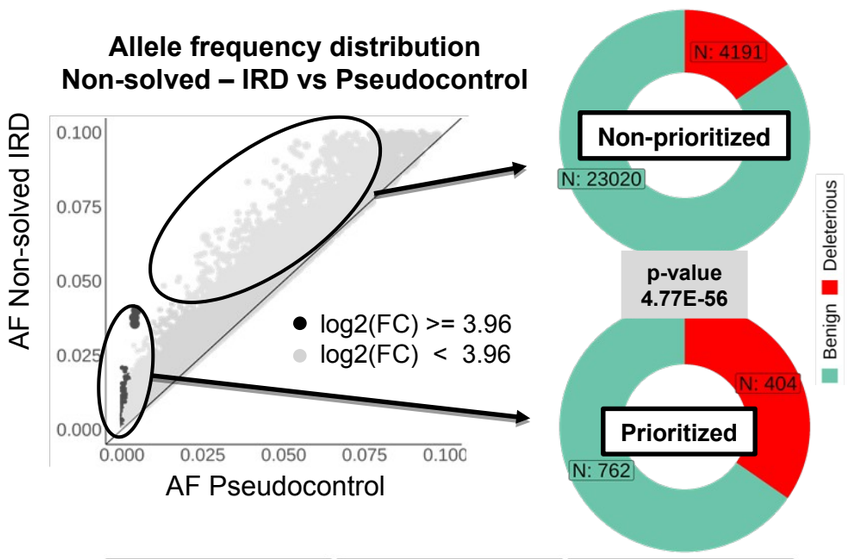
- **IRD** – 244 genes **(IRD genes)**
- **Other eye related diseases (non-IRD) HP:0000478** – 1542 genes **(OERD genes)**
- **Non related diseases genes.** Rest of genes covered in TSO/CES – 3260 genes **(NRD genes)**

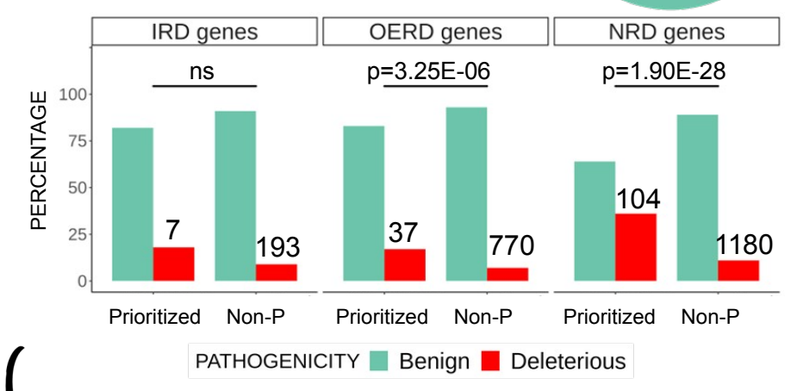
OBJECTIVE : Build a **local allele frequency database** and build **methods to prioritize variants and genes** in IRD non-solved cases.

RESULTS

Variant prioritization

AIM: Extract variants more frequent in IRD cases compared with a set of pseudocontrols.





Review of IRD non-solved cases with prioritized variants

- 4 cases partially solved
- 1 case solved

Applying variant prioritization methodology to VUS reclassification

VUS-ID	log2FC	ACMG + PS4
1	FC >= 1.5	VUS
2	FC >= 1.5	LP
3	FC >= 1.5	LP
4	FC >= 1.5	LP
5	FC >= 1.5	LP
6	FC >= 2.5	LP
7	FC >= 2.5	LP
8	FC >= 2.5	LP
9	FC >= 2.5	LP
10	FC >= 2.5	LP
11	FC >= 2.5	LP

10 / 11 VUS reclassified to Likely pathogenic (LP)

Gene prioritization

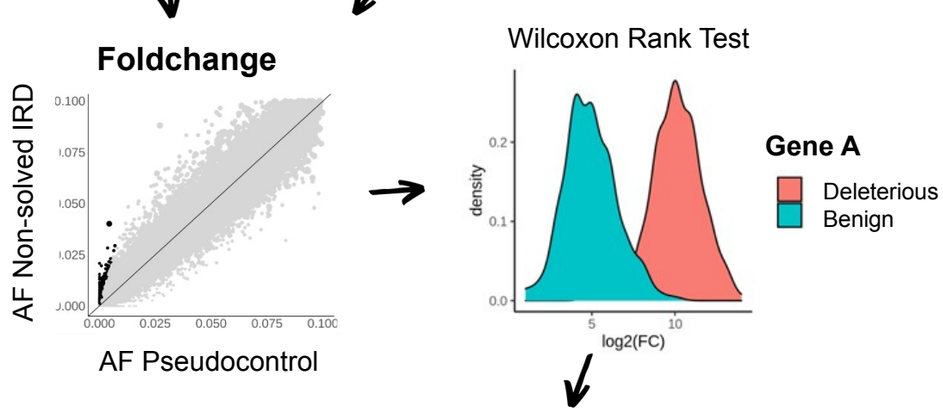
AIM: Extract genes with a higher accumulated pathogenicity in IRD non-solved cases weighted by benign variants.

Gene

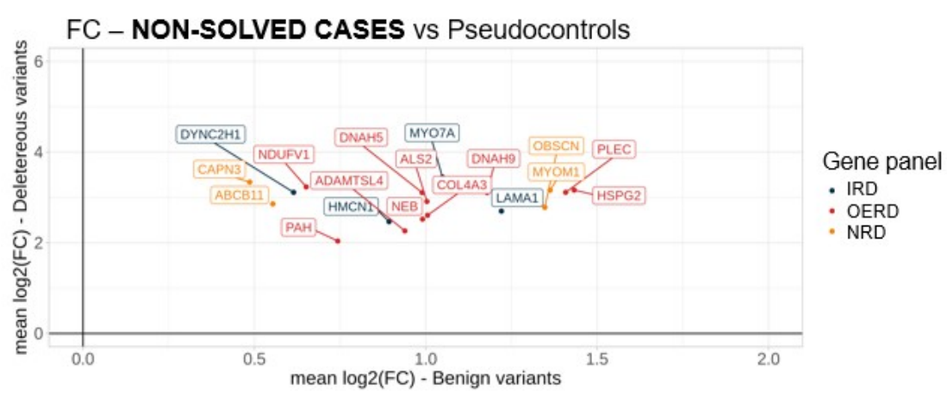
Deleterious variants
- ClinVar Pathogenic
- ClinVar Likely Pathogenic
- CADD_PHRED >= 30

Benign variants
- ClinVar Benign
- ClinVar Likely benign

For each **GENE** in our database



Prioritized genes in IRD NonSolved cases



We identified 18 genes with a higher accumulated pathogenicity in IRD non-solved cases

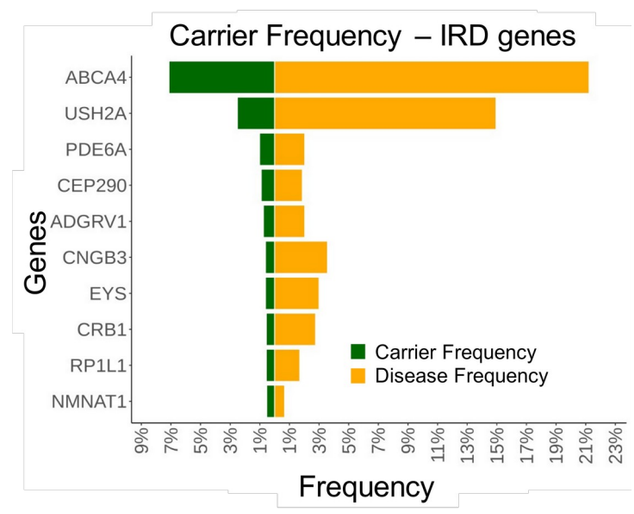
- 4 in IRD panel
- 10 in Eye disease panel (OERD)
- 4 in Other genes panel (NRD)

Review of IRD non-solved cases with variants in prioritized genes

- 2 cases partially solved
- 1 case solved

Carrier frequency

AIM: Calculate carrier frequency for a set of deleterious variants in IRD genes with recessive inheritance pattern.



List of genes with higher carrier frequency

GeneName	AlleleCount	Frequency	Frequency_type
ABCA4	252	7,14	Carrier
USH2A	89	2,52	Carrier
PDE6A	36	1,02	Carrier
CEP290	32	0,91	Carrier
ADGRV1	27	0,76	Carrier
CNGB3	22	0,62	Carrier
EYS	22	0,62	Carrier
CRB1	20	0,57	Carrier
RP1L1	20	0,57	Carrier
NMNAT1	19	0,54	Carrier
ABCA4	375	21,23	IRD
USH2A	264	14,95	IRD
PDE6A	36	2,04	IRD
CEP290	33	1,87	IRD
ADGRV1	36	2,04	IRD
CNGB3	63	3,57	IRD
EYS	53	3,00	IRD
CRB1	49	2,77	IRD
RP1L1	30	1,70	IRD
NMNAT1	12	0,68	IRD

CONCLUSIONS

Cohort-specific allelic frequency database

- This resource is useful for filtering variants by frequency in "local" population.
- Aggregated genomic data can help to develop methods for variants and genes prioritization in IRD non-solved cases.
- Allelic frequency databases can be used to calculate carrier frequency.
- This type of resources can return value to sequenced data.

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